

05 ДЗ - Работа с движками семейства MergeTree.

Собираем Dockerfile.

clickhouse-25.2.1.Dockerfile:

```
FROM clickhouse/clickhouse-server:25.2.1
MAINTAINER Maksim Kulikov <max.uoles@rambler.ru>

RUN apt-get update -y --fix-missing
RUN DEBIAN_FRONTEND=noninteractive apt-get -yq upgrade
RUN apt-get install nano mc python3 pip kafkacat -y
RUN pip install clickhouse_driver

EXPOSE 8123 9000

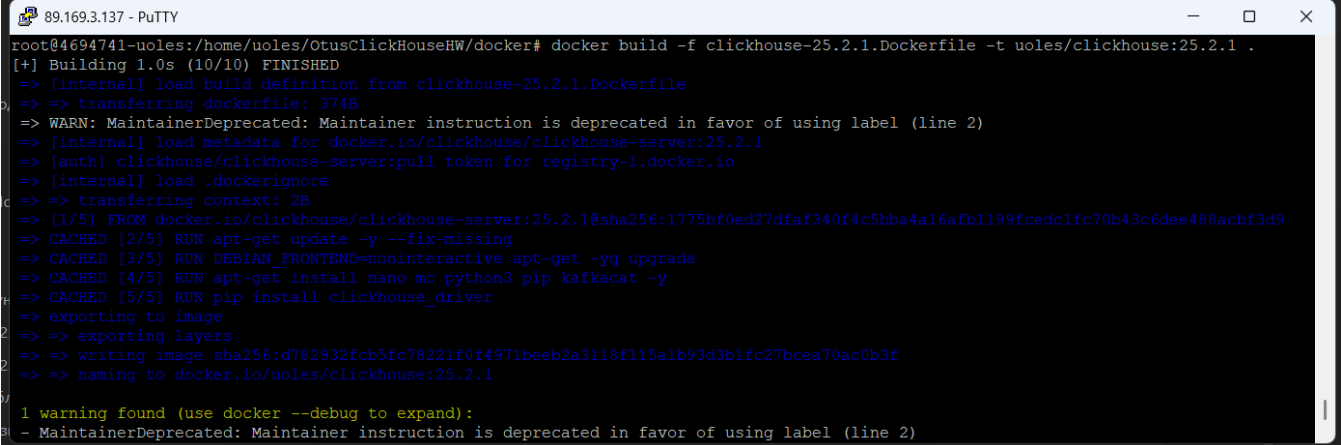
ENTRYPOINT ["/entrypoint.sh"]
```

Собираем образ командой:

```
docker build -f clickhouse-25.2.1.Dockerfile -t uoles/clickhouse:25.2.1 .
```

Запускаем:

```
docker run -d \
-p 18123:8123 \
-p 19000:9000 \
-e CLICKHOUSE_DB=my_database \
-e CLICKHOUSE_USER=username \
-e CLICKHOUSE_DEFAULT_ACCESS_MANAGEMENT=1 \
-e CLICKHOUSE_PASSWORD=password \
--name clickhouse-server \
--ulimit nofile=262144:262144 \
uoles/clickhouse:25.2.1
```



```
89.169.3.137 - PuTTY
root@4694741-uoles:/home/uoles/OtusClickHouseHW/docker# docker build -f clickhouse-25.2.1.Dockerfile -t uoles/clickhouse:25.2.1 .
[+] Building 1.0s (10/10) FINISHED
=> [internal] load build definition from clickhouse-25.2.1.Dockerfile
=> => transferring dockerfile: 374B
=> WARN: MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)
=> [internal] load metadata for docker.io/clickhouse/clickhouse-server:25.2.1
=> [auth] clickhouse/clickhouse-server:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/clickhouse/clickhouse-server:25.2.1@sha256:1775bf0ed27dfaf340f4c5bba4a16afb1199fcedc1fc70b43c6dee488acbf3d9
=> CACHED [2/5] RUN apt-get update -y --fix-missing
=> CACHED [3/5] RUN DEBIAN_FRONTEND=noninteractive apt-get -yq upgrade
=> CACHED [4/5] RUN apt-get install nano mc python3 pip kafkacat -y
=> CACHED [5/5] RUN pip install clickhouse_driver
=> exporting to image
=> => exporting layers
=> => writing image sha256:d782932fcb5fc78221f0f4971beeb2a3118f115alb93d3b1fc27bcea70ac0b3f
=> => naming to docker.io/uoles/clickhouse:25.2.1

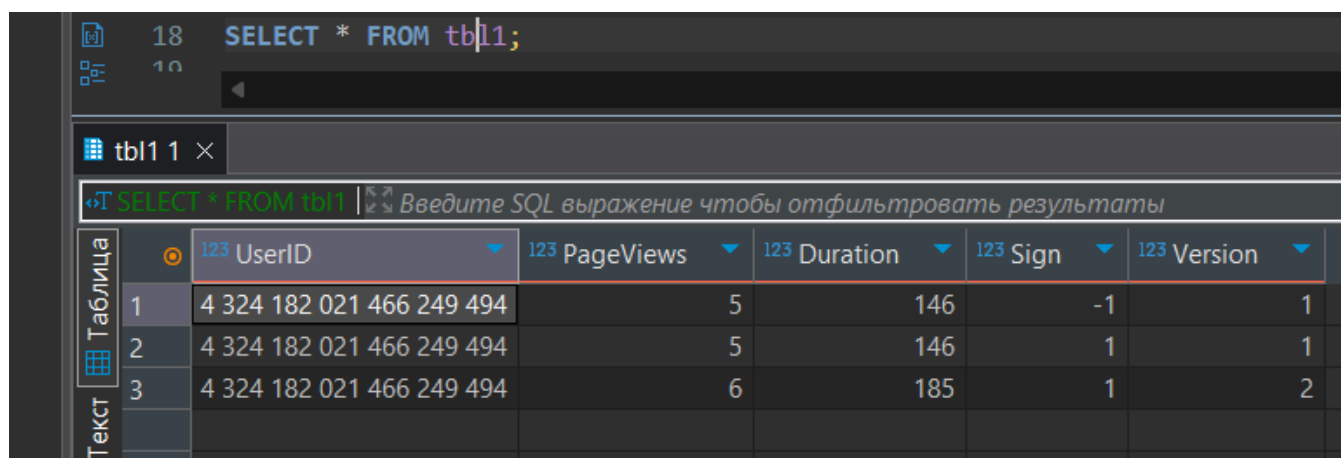
1 warning found (use docker --debug to expand):
- MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)
```

Выполнение заданий.

1. tbl1 с типом VersionedCollapsingMergeTree(Sign, Version) – идет схлопывание записей по значению в поле sign и оставляет запись с наибольшей версией.

```
CREATE TABLE tbl1
(
    UserID UInt64,
    PageViews UInt8,
    Duration UInt8,
    Sign Int8,
    Version UInt8
)
ENGINE = VersionedCollapsingMergeTree(Sign, Version)
ORDER BY UserID;

INSERT INTO tbl1 VALUES (4324182021466249494, 5, 146, -1, 1);
INSERT INTO tbl1 VALUES (4324182021466249494, 5, 146, 1, 1),(4324182021466249494, 6, 185, 1, 2);
```

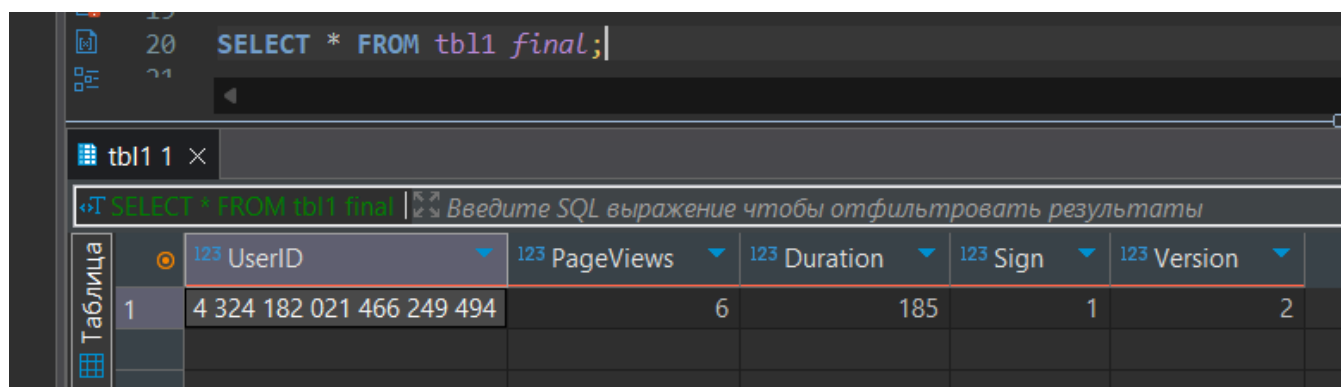


18 SELECT * FROM tbl1;

tbl1 1 x

SELECT * FROM tbl1 | Введите SQL выражение чтобы отфильтровать результаты

Таблица	123 UserID	123 PageViews	123 Duration	123 Sign	123 Version
1	4 324 182 021 466 249 494	5	146	-1	1
2	4 324 182 021 466 249 494	5	146	1	1
3	4 324 182 021 466 249 494	6	185	1	2



20 SELECT * FROM tbl1 final;

tbl1 1 x

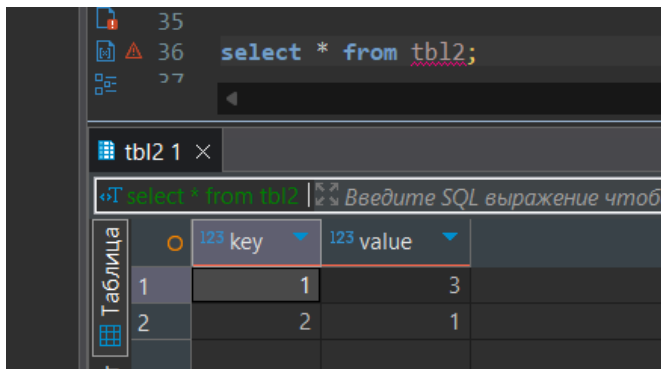
SELECT * FROM tbl1 final | Введите SQL выражение чтобы отфильтровать результаты

Таблица	123 UserID	123 PageViews	123 Duration	123 Sign	123 Version
1	4 324 182 021 466 249 494	6	185	1	2

2. tbl2 с типом SummingMergeTree(value) – идет суммирование значений в поле value по одинаковому ключу – полю key.

```
CREATE TABLE tbl2
(
    key UInt32,
    value UInt32
)
ENGINE = SummingMergeTree(value)
ORDER BY key;

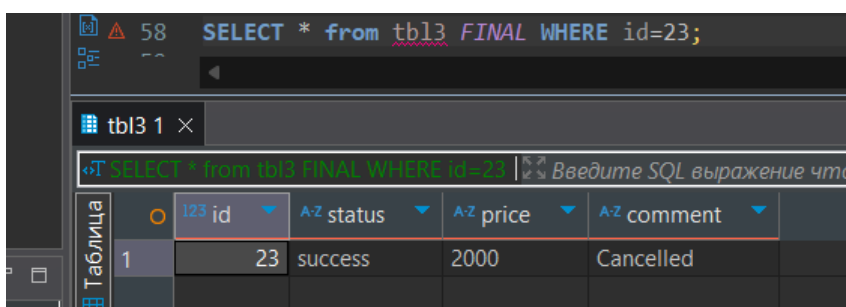
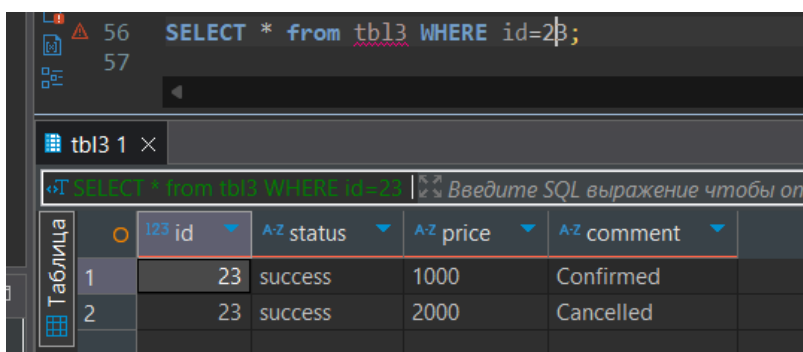
INSERT INTO tbl2 Values(1,1),(1,2),(2,1);
```



3. tbl3 с типом ReplacingMergeTree() – оставляет более свежую запись (с наибольшей версией).

```
CREATE TABLE tbl3
(
  `id` Int32,
  `status` String,
  `price` String,
  `comment` String
)
ENGINE = ReplacingMergeTree()
PRIMARY KEY (id)
ORDER BY (id, status);

INSERT INTO tbl3 VALUES (23, 'success', '1000', 'Confirmed');
INSERT INTO tbl3 VALUES (23, 'success', '2000', 'Cancelled');
```



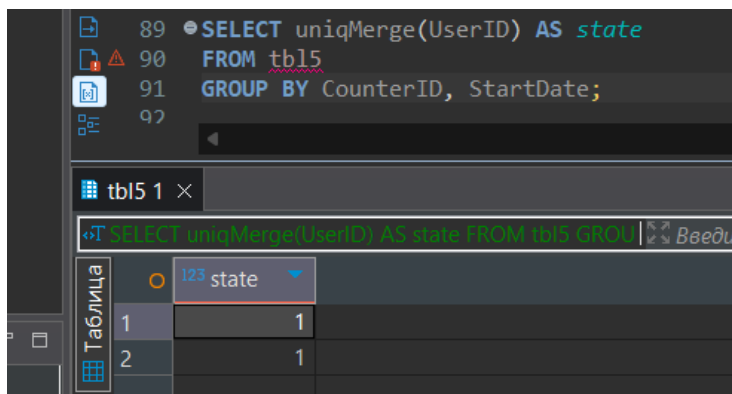
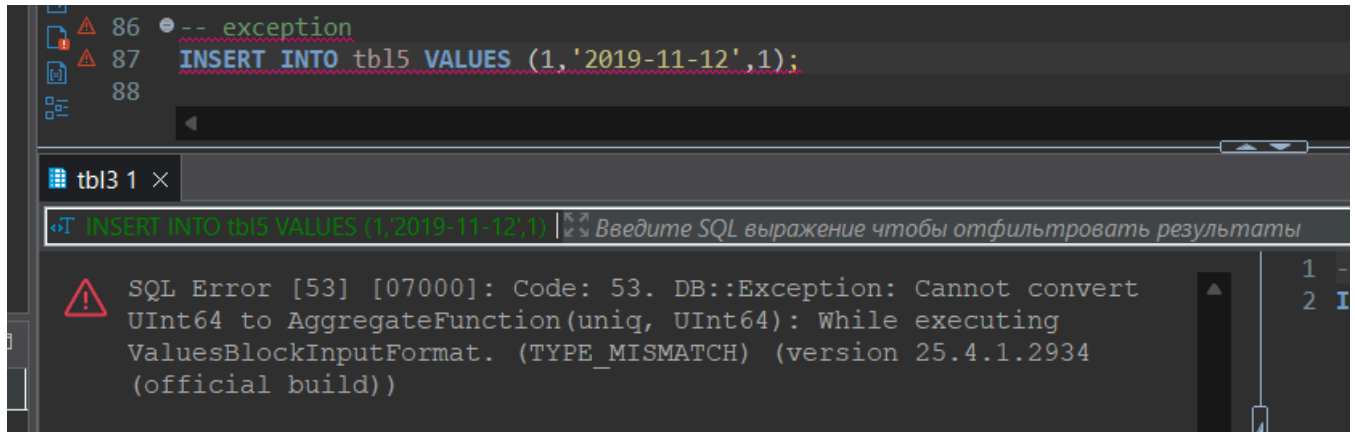
4. tbl4 с типом MergeTree() и tbl5 с типом AggregatingMergeTree() - tbl4 обычная MergeTree, tbl5 в поле UserID хранит значение агрегатной функции.

```
CREATE TABLE tbl4
(
  CounterID UInt8,
  StartDate Date,
  UserID UInt64
) ENGINE = MergeTree()
PARTITION BY toYYYYMM(StartDate)
ORDER BY (CounterID, StartDate);

INSERT INTO tbl4 VALUES(0, '2019-11-11', 1);
INSERT INTO tbl4 VALUES(1, '2019-11-12', 1);
```

```
CREATE TABLE tbl5
(
  CounterID UInt8,
  StartDate Date,
  UserID AggregateFunction(uniq, UInt64)
) ENGINE = AggregatingMergeTree()
PARTITION BY toYYYYMM(StartDate)
ORDER BY (CounterID, StartDate);

INSERT INTO tbl5
select CounterID, StartDate, uniqState(UserID)
from tbl4
group by CounterID, StartDate;
```



- tbl6 с типом CollapsingMergeTree(sign) – удаляет строки по полю Sign для одного значения ключа id.

```
CREATE TABLE tbl6
(
  `id` Int32,
  `status` String,
  `price` String,
  `comment` String,
  `sign` Int8
)
ENGINE = CollapsingMergeTree(sign)
PRIMARY KEY (id)
ORDER BY (id, status);

INSERT INTO tbl6 VALUES (23, 'success', '1000', 'Confirmed', 1);
INSERT INTO tbl6 VALUES (23, 'success', '1000', 'Confirmed', -1), (23, 'success', '2000', 'Cancelled', 1);
```

111

112

SELECT * FROM tbl6;

tbl6 1

SELECT * FROM tbl6

Введите SQL выражение чтобы отфильтровать результа

Таблица

123 id

A-Z status

A-Z price

A-Z comment

123 sign

1

23

success

1000

Confirmed

1

2

23

success

1000

Confirmed

-1

3

23

success

2000

Cancelled

1

113

SELECT * FROM tbl6 FINAL;

114

tbl6 1

SELECT * FROM tbl6 FINAL

Введите SQL выражение чтобы отфильтровать результа

Таблица	123 id	A-Z status	A-Z price	A-Z comment	123 sign
1	23	success	2000	Cancelled	1